Chapter 10 Intro to Networking Exam

Candidate: Dunkan Gibson (dunkan.gibson)
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Score: 93% Passing Score: 80%

Question 1: ✔ Correct

Your wireless network consists of multiple 802.11n access points that are configured as follows:

- SSID (hidden): CorpNet
- Security: WPA2-PSK using AES
- Frequency: 5.7 GHz
- Bandwidth per channel: 20 MHz

This network is required to support an ever-increasing number of devices. To ensure there is sufficient capacity, you want to maximize the available network bandwidth.

What should you do?

- ➡ Double the bandwidth assigned per channel to 40 MHz.
- Upgrade to 802.11a access points.
- Implement antenna diversity.
- Switch to TKIP encryption.

EXPLANATION

802.11n access points provide an option to allocate double the bandwidth per channel (increasing it to 40 MHz), which results in double the data rate.

The 802.11a standard only supports data rates up to 54 Mbps, which is much slower than 802.11n. Antenna diversity implements multiple antennas to combat multipath interference and improve the reliability of a wireless link; however, it does not increase throughput. TKIP is considered less secure than AES and does not increase the throughput of a wireless network.

REFERENCES

LabSim for Network Pro, Section 10.4.

Question 2: ✔ Correct

You need to place a wireless access point in your two-story building. While trying avoid interference, which of the following is the best location for the access point?

- ➡ On the top floor
- In the kitchen area
- Near the backup generators
- In the basement

EXPLANATION

In general, place access points high up to avoid interference problems caused by going through building foundations.

Do not place the access point next to sources of interference such as other wireless transmitting devices (cordless phones or microwaves) or other sources of interference (motors or generators).

REFERENCES

LabSim for Network Pro, Section 10.7.

Question 3: ✔ Correct

Which of the following measures will make your wireless network invisible to the casual attacker performing war driving?

- Change the default SSID
- Implement WPA2 Personal
- ➡ Disable SSID broadcast
- Use a form of authentication other than open authentication

EXPLANATION

Wireless access points are transceivers that transmit and receive information on a wireless network. Each access point has a service set ID (SSID) that identifies the wireless network. By default, access points broadcast the SSID to announce their presence and make it easy for clients to find and connect to the wireless network. Turn off the SSID broadcast to keep a wireless 802.11x network from being automatically discovered. When SSID broadcasting is turned off, users must know the SSID to connect to the wireless network. This helps to prevent casual attackers from connecting to the network, but any serious hacker with the right tools can still connect to the wireless network.

Using authentication with WPA or WPA2 help prevent attackers from connecting to your wireless network, but does not hide the network. Changing the default SSID to a different value does not disable the SSID broadcast.

REFERENCES

LabSim for Network Pro, Section 10.6.

Question 4: ✔ Correct

Match the wireless signaling method on the left with its definition on the right. (Not all of the signaling methods match a definition.)

Uses a narrow frequency band and hops data signals in a predictable sequence

✔ FHSS

Breaks data into pieces and sends the pieces across multiple frequencies in a defined range.

✔ DSSS

Breaks data into very small data streams in order to send the information across long distances.

✔ OFDM

EXPLANATION

The following table describes the most common signaling methods used by wireless networks.

Method	Description
Frequency Hopping Spread Spectrum (FHSS)	FHSS uses a narrow frequency band and hops data signals in a predictable sequence from frequency to frequency over a wide band of frequencies.
Direct-Sequence Spread Spectrum (DSSS)	DSSS uses a transmitter that breaks data into pieces and sends the pieces across multiple frequencies in a defined range. DSSS is more susceptible to interference and less secure then FHSS.
Orthogonal Frequency-Division Multiplexing (OFDM)	OFDM breaks data into very small data streams in order to send the information across long distances where environmental obstacles may be an issue.

REFERENCES

LabSim for Network Pro, Section 10.1.

Question 5: ✔ Correct

Which data transmission rate is defined by the IEEE 802.11b wireless standard?

- 2 Mbps
- 10 Mbps
- 11 Mbps
- 56 Mbps

EXPLANATION

The IEEE 802.11b standard defines wireless transmission rates up to 11 Mbps. Wireless network interface cards and wireless access points (also called wireless hubs or wireless routers) will automatically negotiate the best transmission speed up to 11 Mbps based on current network traffic load and the quality of the wireless connection between the client and access point. The wireless communications are affected by distance, dense physical obstructions, and other electromagnetic interference producing devices.

The IEEE 802.11a standard defines wireless transmission rates up to 2 Mbps. The IEEE 802.11g standard defines wireless transmission rates up to 56 Mbps. The IEEE 802.3 standard defines Ethernet 10baseT cable based transmissions of 10 Mbps.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 6: ✔ Correct

All of the 802.11 standards for wireless networking support which type of communication path sharing technology?

- Token passing
- Polling
- ➡ CSMA/CA
- CSMA/CD

EXPLANATION

802.11x standards for wireless networking all support the CSMA/CA (carrier sense multiple access with collision avoidance) type of communication path sharing technology. This CSMA/CA allows multiple baseband clients to share the same communication medium. CSMA/CA works as follows:

1. The system asks for permission to transmit.
2. A designated authority (such as a hub, router, or access point), grants access when the communication medium is free.
3. The system transmits data and waits for an ACK (acknowledgment).
4. If no ACK is received, the data is retransmitted.

Polling is a mechanism where one system is labeled as the primary system. The primary system polls each secondary system in turn to inquire whether they have data to transmit. Token passing is a mechanism that uses a digital pass card. Only the system holding the token is allowed to communicate. CSMA/CD is the technology used by Ethernet. CSMA/CD works as follows:

1. The system listens for traffic. If the line is clear, the system begins transmitting.
2. During the transmission, the system listens for collisions.
3. If no collisions are detected, the communication succeeds. If collisions are detected, an interrupt jam signal is broadcast to stop all transmissions. Each system waits a random amount of time before starting over at step 1.

REFERENCES

LabSim for Network Pro, Section 10.1.

Question 7: ✔ Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You completed the lab correctly.

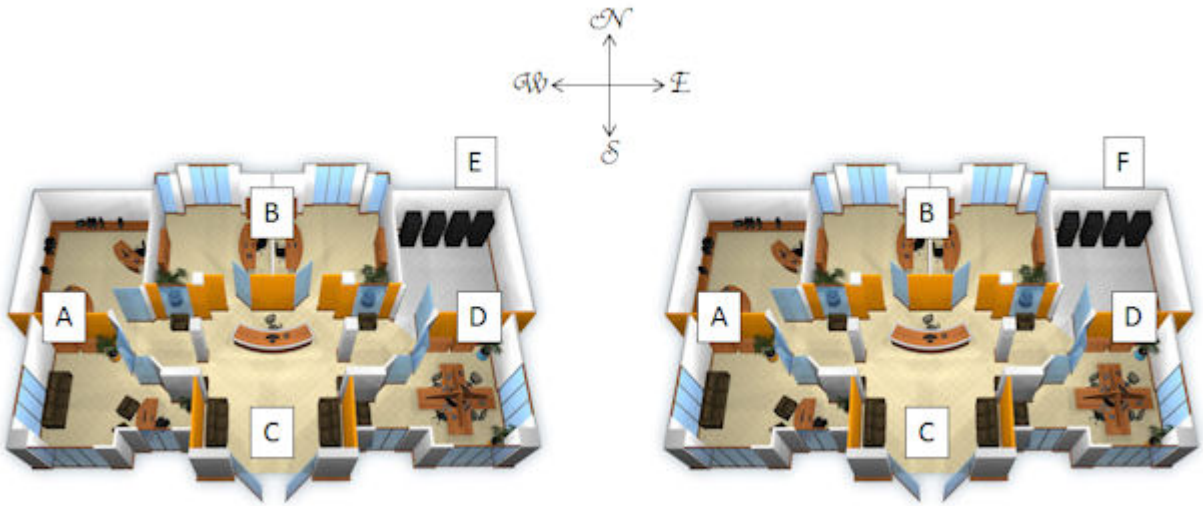
View Lab Report

Question 8: ✔ Correct

Your consulting firm has been hired by a small business to implement a wireless network. The company leases two office suites within a business park approximately 200m apart, as shown below. The objectives of the implementation are as follows:

- Create a secure wireless network that doesn't emanate beyond each office space by implementing access points in locations A-D in each building.
- Connect the wireless networks at each office together with a secure outdoor wireless link using locations E and F.

Drag the antenna type from the list on the left to the appropriate location on the right. Each antenna type can be used more than once or not at all.



A
✔ Normal gain directional antenna

B
✔ Normal gain east directional antenna

C
✔ Normal gain south directional antenna

D
✔ Normal gain north directional antenna

E
✔ High-gain directional west antenna aimed

F
✔ High-gain directional

antenna
aimed
west

EXPLANATION

In this scenario, directional antennae can be implemented along the exterior walls that are aimed inward. This reduces signal emanation outside the organization. To reduce coverage gaps, you could also implement an omni-directional antenna in the center of each office complex with the power turned to prevent signal emanation.

To establish a wireless link between offices, you can implement high-gain directional parabolic antennae at each site. Because this radio signal will be transmitted outdoors, you will need to use the strongest encryption available on the link.

REFERENCES

LabSim for Network Pro, Section 10.6.

Question 9: ✔ Correct

Which of the following locations creates the greatest amount of interference for a wireless access point? (Select two.)

- [ ] Near DCHP servers
- [x] Near backup generators
- [ ] In the top floor of a two-story building
- [x] Near cordless phones

EXPLANATION

Other wireless transmitting devices (such as cordless phones or microwaves) and generators cause interference for wireless access points.

In general, place access points higher up to avoid interference problems caused by going through building foundations. DHCP servers provide IP information for clients and do not cause interference.

REFERENCES

LabSim for Network Pro, Section 10.6.

Question 10: ✔ Correct

Match each type of access point on the left with the wireless network architecture where it is commonly used on the right. Each type of access point may be used once, more than once, or not at all.

Independent access point infrastructure

✔ Intelligent AP

Hub-and-spoke infrastructure

✔ Lightweight AP

Distributed wireless mesh infrastructure

✔ Intelligent AP

EXPLANATION

Different types of access points are used in different wireless network architectures, as described in the following table:

Architecture	Description
Independent Access Points	In the early days of wireless networking, large organizations implemented independent access points throughout their facilities. Each AP stood alone, providing separate wireless networks using its own independent configuration.
Hub-and-Spoke Infrastructure	In a hub-and-spoke configuration, a wireless controller is connected to all access points using wired links. The individual access points contain very little embedded intelligence and are sometimes referred to as lightweight access points (LWAPs).
Distributed Wireless Mesh Infrastructure	Newer wireless networks can be deployed using a distributed wireless mesh architecture. These networks still use a controller, but they move some of the network intelligence from the controller out to the individual access points. In this configuration, the controller is no longer a bottleneck. The APs are smart enough to communicate directly with each other to create more efficient data paths for network traffic.

REFERENCES

LabSim for Network Pro, Section 10.5.

Question 11: ✔ Correct

You are designing an update to your client's wireless network. The existing wireless network uses 802.11b equipment; which your client complains runs too slowly. She wants to upgrade the network to run up to 600 Mbps.

Due to budget constraints, your client wants to upgrade only the wireless access points in the network this year. Next year, she will upgrade the wireless network boards in her users' workstations. She has also indicated that the system must continue to function during the transition period.

Which 802.11 standard will work best in this situation?

- 802.11b
- 802.11d
- ➡ 802.11n
- 802.11c
- 802.11a

EXPLANATION

802.11n is the best choice for this client and provides up to 600 Mbps. With 802.11n, you may have a single device that uses multiple radios, one that can operate at one frequency and another that can operate on a different frequency. Because of this, 802.11n usually allows for compatibility between all 802.11 standards, depending upon the specific implementation.

While 802.11g is compatible with 802.11b, it only provides up to 54 Mbps.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 12: ✔ Correct

You are designing a wireless network for a client. Your client needs the network to support a data rate of at least 54 Mbps. In addition, the client already has a wireless telephone system installed that operates at 2.4 GHz.

Which 802.11 standards will work best in this situation? (Select two.)

- [ ] 802.11b
- [x] 802.11a
- [ ] 802.11d
- [x] 802.11n
- [ ] 802.11g

EXPLANATION

802.11a or 802.11n are the best choices for this client.

While both 802.11a and 802.11g each operate at 54 Mbps, 802.11g operates in the 2.4 GHz to 2.4835 GHz range—which will cause interference with the client's wireless phone system. 802.11a and 802.11n, on the other hand, operate in the 5.725 GHz to 5.850 GHz frequency range. This won't interfere with the phone system. 802.11n can operate at speeds up to 600 Mbps.

REFERENCES

LabSim for Network Pro, Section 10.4.

Question 13: ✔ Correct

You are configuring a wireless network with two wireless access points. Both access points connect to the same wired network. You want wireless users to be able to connect to either access point and have the ability to roam between the two access points.

How should you configure the access points?

- Different SSID, different channel
- ➡ Same SSID, different channel
- Same SSID, same channel
- Different SSID, same channel

EXPLANATION

When you configure multiple access points as part of the same extended service set (ESS), configure both access points with the same service set identifier (SSID). The SSID is like a network name and groups wireless devices together into the same logical network. All devices, including wireless clients, use the same SSID.

Wireless access points that are in the same area should use different channels. If the channels are the same or overlap, devices connected to one access point might interfere with devices connected to the other access point in locations where the signal overlaps.

REFERENCES

LabSim for Network Pro, Section 10.3.

Question 14: ✔ Correct

Your organization uses an 802.11b wireless network. Recently, other tenants installed the following equipment in your building:

- A wireless television distribution system running at 2.4 GHz
- A wireless phone system running at 5.8 GHz
- A wireless phone system running at 900 MHz
- An 802.11a wireless network running in the 5.725 - 5.850 GHz frequency range
- An 802.11j wireless network running in the 4.9 - 5.0 GHz frequency range

Since this equipment was installed, your wireless network has been experiencing significant interference. Which system is to blame?

- The 900 MHz wireless phone system
- ➡ The wireless TV system
- The 802.11j wireless network
- The 802.11a wireless network
- The 5.8 GHz wireless phone system

EXPLANATION

Because the 802.11b standard operates within the 2.4 GHz to 2.4835 GHz radio frequency range, the most likely culprit is the wireless TV distribution system.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 15: ✔ Correct

After installing a new 2.4Ghz cordless phone system in your office, you notice that wireless network performance is adversely affected. Which of the following wireless networking standards are you most likely using? (Select two.)

- [ ] Bluetooth
- [x] 802.11g
- [ ] 802.11a
- [x] 802.11b

EXPLANATION

Both the 802.11b and 802.11g wireless networking standards use the 2.4Ghz frequency range. A cordless phone system on the same frequency range may affect the performance of the wireless network.

802.11a uses the 5Ghz frequency range, so it would not be affected by a cordless phone system that uses the 2.4Ghz frequency range. Bluetooth does use the 2.4Ghz frequency range, but is used more widely as a mechanism to connect consumer electronic devices like personal digital assistants (PDAs), cameras, and phones, rather than as a wireless local area networking (LAN) method.

REFERENCES

LabSim for Network Pro, Section 10.7.

Question 16: ✔ Correct

Your wireless network consists of multiple 802.11n access points that are configured as follows:

- SSID (hidden): CorpNet
- Security: WPA2-PSK using AES
- Frequency: 5.75 GHz
- Bandwidth per channel: 40 MHz

Because of the unique construction of your organization's facility, there are many locations that do not have a clear line of sight between network clients and access points. As a result, radio signals are reflected along multiple paths before finally being received. The result is distorted signals that interfere with each other.

What should you do?

- Reduce the power of the access point radio signals.
- ➡ Implement antenna diversity.
- Install directional access points.
- Switch to RADIUS authentication for wireless clients.

EXPLANATION

Antenna diversity implements two or more radio antennae to improve the quality and reliability of a wireless link. In environments where there is no clear line of sight between transmitter and receiver, the radio signal is reflected along multiple paths before finally being received. This can introduce phase shifts, time delays, attenuation, and distortion that interfere with each another on the receiving antenna.

You can rectify the situation by implementing antenna diversity two ways:

- *Spatial diversity* uses multiple antennas that are physically separated from one another.
- *Pattern diversity* uses two or more co-located antennas with different radiation patterns.

Using a RADIUS authentication solution increases wireless network security, but it doesn't address the issue of multipath interference. Reducing radio power could help solve multipath interference issues in some situations, but it may make it worse in others. This is also true of directional access points.

REFERENCES

LabSim for Network Pro, Section 10.7.

Question 17: ✔ Correct

Which technologies are used by the 802.11ac standard to increase network bandwidth? (Select two.)

- [ ] Data compression
- [x] Eight MIMO radio streams
- [ ] Four MIMO radio streams
- [x] 160 MHz bonded channels
- [ ] 40 MHz bonded channels

EXPLANATION

To increase network bandwidth, the 802.11ac standard uses:

- Eight MIMO radio streams
- 160 MHz wide bonded channels

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 18: ✔ Correct

Which IEEE standard describes wireless communication?

- (●) 802.11b
- ( ) 802.2
- ( ) 802.7b
- ( ) 802.5
- ( ) 802.3

EXPLANATION

One IEEE standard for wireless is 802.11b.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 19: ✔ Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You completed the lab correctly.

View Lab Report

Question 20: ✕ Incorrect

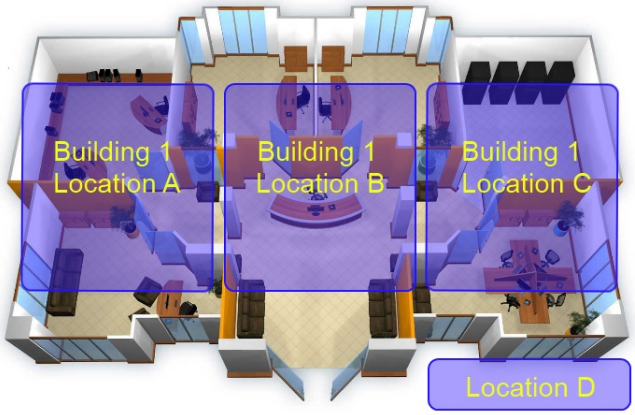
You are building a wireless network within and between two buildings. The buildings are separated by more than 3000 feet. The wireless network should meet the following requirements:

- Wireless data within Building 1 should be protected with the highest degree of security.
- Wireless data within Building 2 should be accessible and permitted by any wireless client.
- Wireless signals between Buildings 1 and 2 should be protected with the highest degree of security.
- Wireless signals within Buildings 1 and 2 should cover the whole structure, but not extend to the outside.

For each location on the image below, you need to select the following:

- Antenna option
- Security option

Drag the items from the list on the left to the location identifier on the right. Items may be used more than once. Not all items will be used.



Building 1 - Location A

✔ WPA2 with CCMP

✔ Right-facing directional antenna

Building 1 - Location B

✔ WPA2 with CCMP

✔ Omni-directional antenna

Building 1 - Location C

✔ WPA2 with CCMP

✔ Left-facing directional antenna

Building 1 - Location D

✔ WPA2 with CCMP

✔ Right-facing high-gain directional

Building 2 - Location A

~~Left-facing~~ Omni-directional

~~WPA2 with~~ WEP with open

~~high-gain directional antenna~~	antenna	~~CCMP~~	authentication

Building 2 - Location B

~~Omni-directional antenna~~	Left-facing high-gain directional antenna	~~WEP with open authentication~~	WPA2 with CCMP

EXPLANATION

To answer this question correctly, you should choose the following:

Building 1 - Location A = Right-facing directional antenna, WPA2- CCMP
Building 1 - Location B = Omni-directional antenna, WPA2- CCMP
Building 1 - Location C = Left-facing directional antenna, WPA2- CCMP
Building 1 - Location D = Right-facing parabolic antenna, WPA2- CCMP
Building 2 - Location A = Omni-directional antenna, WEP with open authentication
Building 2 - Location B = Left-facing parabolic antenna, WPA2- CCMP

Be aware of the following types of antennas:

- Directional antenna:
  - Creates a narrow, focused signal in a particular direction.
  - Focuses the signal to provide greater signal strength, thus increasing the transmission distance.
  - Provides a stronger point-to-point connection, better equipping them to handle obstacles. A *parabolic* directional antenna is highly focused, sending and receiving signals in far greater distances than achieved with a typical directional antenna.
- Omni-directional antenna:
  - Disperses the RF wave in an equal 360-degree pattern.
  - Provides access to many clients in a radius.

Be aware of the following types of security:

- Wired Equivalent Privacy (WEP) is an optional component of the 802.11 specifications, but is easily broken. When using WEP, use open authentication.
- Wi-Fi Protected Access 2 (WPA2) resolves the weaknesses inherent in WEP. WPA2 uses counter mode with the CBC-MAC protocol (CCMP), also known as AES-CCMP. Note that WPA2 does not use TKIP.

REFERENCES

LabSim for Network Pro, Section 10.4.

Question 21: ✔ Correct

What is the **least** secure place to locate an access point with an omni-directional antenna when creating a wireless cell?

- Above the 3rd floor
- ➡ Near a window
- In common or community work areas
- In the center of the building

EXPLANATION

The least secure location for a wireless cell access point is against a perimeter wall. Placement near a window would be the worst option from this list of selections.

For the best security, access points that use directional antennae would be a more appropriate choice for placement near an exterior wall. This placement reduces the likelihood that the wireless cell's access radius will extend outside of the physical borders of your environment. It is important to place wireless cell access points where they are needed, such as in a common or community work area.

REFERENCES

LabSim for Network Pro, Section 10.6.

Question 22: ✔ Correct

Match the wireless networking term or concept on the left with its appropriate description on the right. Each term may be used once, more than once, or not at all.

Moving an wireless device between access points within the same wireless network.

✔ Roaming

Used by Cisco wireless equipment to route frames back and forth between the wireless network and the wired LAN.

✔ LWAPP

Specifies the number of clients that utilize the wireless network.

✔ Device density

Automatically partitions a single broadcast domain into multiple VLANs.

✔ VLAN pooling

Graphically displays wireless signal strength within an area.

✔ Heat map

Connects two wired networks over a Wi-Fi network.

✔ Wireless bridge

Identifies relative strength of a radio signal at the receiver.

✔ Heat map

The number of useful bits delivered from sender to receiver within a specified amount of time.

✔ Goodput

EXPLANATION

You should be familiar with the following wireless networking terms and concepts:

- Device density specifies the number of clients that utilize the wireless network.
- Roaming is moving an wireless device between access points within the same wireless network.
- The Lightweight Access Point Protocol (LWAPP) is used by Cisco wireless equipment to route frames back and forth between the wireless network and the wired LAN.
- VLAN pooling automatically partitions a single broadcast domain into multiple VLANs.
- A wireless bridge connects two wired networks over a Wi-Fi network.
- A heat map graphically displays the relative wireless signal strength within a wireless deployment.
- Goodput refers to the number of useful bits delivered from the sender to the receiver within a specified amount of time.

REFERENCES

LabSim for Network Pro, Section 10.5.

Question 23: ✔ Correct

You have configured a wireless access point to create a small network. For security, you have disabled SSID broadcast.

From a client computer, you try to browse to find the access point. You see some other wireless networks in the area, but cannot see your network.

What should you do?

- Enable the wireless card on the client.
- Set the channel on the client to match the channel used by the access point.
- Decrease the beacon interval on the access point.
- ➡ Configure a profile on the wireless client.

EXPLANATION

When the SSID broadcast is turned off, you must manually configure a profile on the client computer to identify the SSID of the access point. If you disable the SSID broadcast, you must statically configure wireless devices with the SSID before they can connect because they will be unable to dynamically detect the SSID.

A beacon is a frame that the access point sends out periodically. When you turn off SSID broadcast, you prevent the access point from including the SSID in the beacon. On the client, the channel is typically detected automatically and is configured to match the channel used by the access point. In this scenario, the wireless card on the client is already enabled because you can see other wireless networks in the area.

REFERENCES

LabSim for Network Pro, Section 10.3.

Question 24: ✔ Correct

You are implementing a wireless network inside a local office. You require a wireless link to connect a laptop in the administrator's office directly to a system in the sales department. In the default configuration, the wireless AP uses a 360-dispersed RF wave design. After installation, the signal between the two systems is weak, as many obstacles interfere with the signal.

Which of the following strategies could you try to increase signal strength?

- Replace the directional antenna with an omni-directional antenna.
- ➡ Replace the omni-directional antenna with a directional antenna.
- Increase the RF setting on the AP.
- Increase the RF setting on the client system.
- Increase the RF power on the isotropic antenna.

EXPLANATION

A directional antenna is designed to create a narrow, focused signal in a particular direction. This focused signal provides greater signal strength between two points and increases the distance that the signal can travel. Because directional antennas provide a stronger point-to-point connection, they are better equipped to handle obstacles that may be in the way of the signal.

The default antenna used with this configuration is an omni-directional antenna that disperses the RF wave in an equal 360-degree pattern. This antenna is commonly used to provide access to many clients in a radius.

REFERENCES

LabSim for Network Pro, Section 10.7.

Question 25: ✔ Correct

You are designing a wireless network for a client. Your client needs the network to support a data rate of at least 150 Mbps. In addition, the client already has a wireless telephone system installed that operates 2.4 GHz.

Which 802.11 standard will work best in this situation?

- 802.11a
- 802.11g
- 802.11b
- ➡ 802.11n

EXPLANATION

802.11n is the best choice for this client.

802.11b and 802.11g both operate in the 2.4 GHz to 2.4835 GHz range, which will cause interference with the client's wireless phone system. 802.11a operates in the 5.725 GHz to 5.850 GHz frequency range. While this won't interfere with the phone system, its maximum speed is limited to 54 Mbps.

REFERENCES

LabSim for Network Pro, Section 10.4.

Question 26: ✔ Correct

Which IEEE wireless standards specify transmission speeds up to 54 Mbps? (Select two.)

- [ ] 802.11b
- [ ] Bluetooth
- [ ] 802.1x
- [x] 802.11g
- [x] 802.11a

EXPLANATION

Both the 802.11a and the 802.11g wireless standards specify maximum transmission speeds up to 54 Mbps.

Bluetooth is a wireless standard commonly used to connect peripheral devices and operates at 720 Kbps. The 802.11b wireless standard provides transmission speeds of 11 Mbps. 802.1x is a wireless security standard that provides an authentication framework for 802-based networks.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 27: ✔ Correct

Which of the following enterprise wireless deployment models uses access points with enough intelligence to allow the creation of guest WLANs for keeping public wireless traffic separate from private traffic?

- Independent access points
- Hub-and-spoke infrastructure
- Lightweight access point (LWAP) with wireless controller infrastructure
- ➡ Distributed wireless mesh infrastructure

EXPLANATION

A distributed wireless mesh architecture moves some of the network intelligence from the controller out to the individual access points. In this configuration, the controller is no longer a bottleneck. The APs are smart enough to communicate directly with each other to create more efficient data paths for network traffic.

With the hub-and-spoke infrastructure, the individual access points contain very little embedded intelligence and are sometimes referred to as lightweight access points (LWAPs). Independent access points offer limited mobility and require the individual configuration of each AP.

REFERENCES

LabSim for Network Pro, Section 10.5.

Question 28: ✔ Correct

Which of the following do switches and wireless access points use to control access through a device?

- Port number filtering
- ➡ MAC filtering
- IP address filtering
- Session filtering

EXPLANATION

Both switches and wireless access points are Layer 2 devices, meaning they use the MAC address to make forwarding decisions. Both devices typically include some form of security that restricts access based on the MAC address.

Routers and firewalls operate at Layer 3 and can use the IP address or port number for filtering decisions. A circuit-level gateway is a firewall that can make forwarding decisions based on the session information.

REFERENCES

LabSim for Network Pro, Section 10.6.

Question 29: ✔ Correct

Which of the following specifications identify security that can be added to wireless networks? (Select two.)

- [ ] 802.2
- [ ] 802.11a
- [x] 802.11i
- [ ] 802.3
- [ ] 802.5
- [x] 802.1x

EXPLANATION

802.11i defines security for wireless networks. Standards described in 802.11i have been implemented in Wi-Fi Protected Access (WPA) and Wi-Fi Protected Access 2 (WPA2). 802.1x is an authentication protocol that can be used on wireless networks. Use of 802.1x on a wireless network is described in the 802.11i specifications.

802.11a is a wireless standard that describes how wireless communications work. However, the standard does not include security considerations. 802.2 defines standards for data link layer communications. 802.3 is Ethernet, and 802.5 is Token Ring.

REFERENCES

LabSim for Network Pro, Section 10.6.

Question 30: ✔ Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You completed the lab correctly.

View Lab Report

Question 31: ✔ Correct

How many total channels are available for 802.11a wireless networks?

- 3
- 11
- 12
- 23
- 54

EXPLANATION

802.11a wireless uses the 5.75 GHz range which has a total of 23 channels.

802.11b and 802.11g use the 2.4 GHz range which has a total of 11 channels in the US.

REFERENCES

LabSim for Network Pro, Section 10.2.

Question 32: ✔ Correct

An attacker is trying to compromise a wireless network that has been secured using WPA2-PSK and AES. She first tried using AirSnort to capture packets, but found that she couldn't break the encryption.

As an alternative, she used software to configure her laptop to function as an access point. She configured the fake access point with the same SSID as the wireless network she is trying to break into. When wireless clients connect to her access point, she presents them with a web page asking them to enter the WPA2 passphrase. When they do, she then uses it to connect a wireless client to the real access point.

Which attack techniques did the attacker use in this scenario? (Select two.)

- [ ] Smurf
- [x] Pharming
- [ ] Denial of service
- [ ] Man-in-the-middle
- [x] Evil twin

EXPLANATION

The attacker in this scenario used the following attack techniques:

- Evil twin: In this exploit, an attacker near a valid wireless access point installs an access point with the same (or similar) SSID.
- Pharming: In this exploit, the access point is configured to display a bogus web page that prompts for credentials, allowing the attacker to steal those credentials.

Denial of service attacks overload a target system to the point that it can no longer perform its desired function on the network. A man-in-the-middle attack occurs when the attacker gets in between a sender and receiver, posing as the sender to the receiver and as the receiver to the sender. A Smurf attack is a type of denial of service attack that uses spoofed ICMP echo response packets from an amplifier network to overload a target host.

REFERENCES

LabSim for Network Pro, Section 10.6.

Question 33: ✕ Incorrect

To answer this question, complete the lab using the information below. **Launch Lab**

You did not attempt the lab.

Question 34: ✔ Correct

Match each wireless term or concept on the left with its associated description on the right. Each term may be used more than once; not all descriptions have a matching term.

Compares the level of the Wi-Fi signal to the level of background radio signals.

✔ Signal to noise ratio

Checks channel utilization and identifies sources of RF inference.

✔ Spectrum analysis

Identifies how strong a radio signal is at the receiver.

✔ Received signal level

Identifies the number of useful bits delivered from the sender to the receiver.

Causes multiple copies of the same radio signal to be received by the receiving antenna.

✔ Bounce

Degrades wireless network performance.

✔ Device saturation

EXPLANATION

You should be familiar with the following wireless networking concepts and terms:

- *Bounce* can cause multiple copies (one from each path) of the same signal to be received by the receiving antenna.
- *Received Signal Level* (RSL) identifies how strong the radio signal is at the receiver. The closer you are to the transmitter, the stronger the RSL.
- *Signal to Noise Ratio* (SNR) compares the level of the wireless network signal (RSL) to the level of background noise.
- A *spectrum analysis* is used to check channel utilization and to identify sources of RF inference at each location where you plan to deploy an access point.
- *Device saturation* occurs when the wireless network is fully utilized and can no longer support additional wireless clients. Adding more clients can severely degrade network performance.

REFERENCES

LabSim for Network Pro, Section 10.4.

Question 35: ✕ Incorrect

To answer this question, complete the lab using the information below. **Launch Lab**

You did not attempt the lab.

Question 36: ✔ Correct

You are the wireless network administrator for your organization. As the size of the organization has grown, you've decided to upgrade your wireless network to use 802.1x authentication instead of using preshared keys.

To do this, you need to configure a RADIUS server and RADIUS clients. You want the server and the clients to mutually authenticate with each other.

What should you do? (Select two. Each response is a part of the complete solution.)

- [ ] Configure all RADIUS clients with a preshared key.
- [x] Configure all wireless access points with client certificates.
- [x] Configure the RADIUS server with a server certificate.
- [ ] Configure the RADIUS server with a preshared key.
- [ ] Configure all wireless workstations with client certificates.

EXPLANATION

When using 802.1x authentication for wireless networks, a RADIUS server is implemented to centralize authentication. A centralized authentication database is used to allow wireless clients to roam between cells and authenticate to each using the same account information. PKI is required for issuing certificates. At a minimum, the RADIUS server must have a server certificate; however, to support mutual authentication, each RADIUS client must also have a certificate. Remember that each wireless access point in a RADIUS solution is a RADIUS client, not the wireless devices. The wireless access points forward the credentials from wireless devices to the RADIUS server for authentication.

Preshared keys are not used for authentication in an 802.1x solution.

REFERENCES

LabSim for Network Pro, Section 10.6.

Question 37: ✔ Correct

Which wireless networking component is used to connect multiple APs together?

- STA
- ➡ DS
- IBSS
- BSS

EXPLANATION

The distribution system (DS) is the backbone or LAN that connects multiple APs (and BSSs) together. The DS allows wireless clients to communicate with the wired network and with wireless clients in other cells.

An IBSS is a set of STAs configured in ad hoc mode. A BSS, or cell, is the smallest unit of a wireless network. An STA is a wireless NIC in an end device such as a laptop or wireless PDA. The term STA often refers to the device itself, not just the NIC.

REFERENCES

LabSim for Network Pro, Section 10.1.

Question 38: ✔ Correct

Which of the following protocols or mechanisms is used to provide security on a wireless network? (Select three.)

- [ ] RDP
- [x] WPA
- [x] IPsec
- [x] 802.1x

EXPLANATION

Remote Desktop Protocol (RDP) is used by Microsoft Windows Terminal Services applications, such as Remote Desktop. It is not used to provide security on wireless networks.

IPsec is an encryption and authentication mechanism designed to provide security for the TCP/IP protocol suite. It is often used on wireless networks to ensure data integrity and authenticity. Wi-Fi Protected Access (WPA) is a robust security protocol designed to provide additional security to wireless networks. WPA authenticates devices to the wireless network and provides encryption services to protect data as it travels across the wireless network. 802.1x is an authentication mechanism for wireless networks. 802.1x generally uses a Remote Authentication Dial-In User Service (RADIUS) server to authenticate users to the wireless network.

REFERENCES

LabSim for Network Pro, Section 10.6.

Question 39: ✔ Correct

Which protocol is well known for its use in the the home security and home automation industry, uses a mesh topology, makes devices act as repeaters, and has a low data transfer rate?

- ➡ Z-Wave
- Ant+
- NFC
- 802.11 AC

EXPLANATION

The Z-Wave protocol is mostly used in the home security and automation market and uses only a mesh topology. Each device that is added acts as a repeater and increase the strength of the network. Z-Wave has a low data transfer rate.

Ant+ is can be used in a mesh topology, but is generally used to monitor sensor data. NFC is commonly used for mobile pay solutions and connections like blue tooth, but has to be several inches within another device to connect. 802.11 AC is a wireless networking standard that offers high-speed data transfer.

REFERENCES

LabSim for Network Pro, Section 10.4.

Question 40: ✔ Correct

What is the risk associated with smart technology used in networked devices such as smart refrigerators, environmental controls, or industrial equipment?

- ➡ They are vulnerable to exploits due to weaker security.
- They are not physically secure, so they can be reconfigured by anyone who has access to them.
- They are not compatible with routing protocols, which causes a burden on network bandwidth.
- They use redundant paths for fault tolerance that can cause message loops to develop on the network.

EXPLANATION

Devices with embedded smart technology are not designed to be customized or directly configured by system administrators. For example, you cannot install anti-malware software on a smart TV. Because of this weaker security, smart technology devices are vulnerable to exploits and attacks.

REFERENCES

LabSim for Network Pro, Section 10.4.

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